

PLC Control of Gantry Vacuum Loader

6.1 Inputs

The system uses:

- Start push button (I0.0) and Stop push button (I0.1) to enable and disable the automatic cycle.
- Emergency stop (I0.2) to immediately stop the machine and de-energize all outputs.
- Horizontal position sensors (I0.3, I0.4) to detect when the gripper is above the stack or above the conveyor.
- Vertical position sensors (I0.5, I0.6) to detect when the gripper is in the upper or lower position.
- Stack sensor (I1.0) to make sure there are still sheets in the stack.
- Conveyor sensor (I0.7) to detect if there is already a sheet at the end of the conveyor, to avoid collisions and control spacing.
- Vacuum pressure sensor (I1.1) to confirm that the sheet is correctly gripped.

6.2 Outputs

- Horizontal drives (Q0.0, Q0.1) move the gantry horizontally.
- Vertical drives (Q0.3, Q0.2) move the gripper up or down.
- Vacuum valve (Q0.4) grips or releases the sheet.
- Conveyor motor (Q0.5) transports the sheets forward.

6.3 Memory and Control Concept

Internal memory bits M0.1...M0.6 are used as step flags for each stage:

1. Move to stack
2. Lower to pick
3. Grip and raise
4. Move to conveyor
5. Lower and release
6. Raise and return

Each step activates only after the previous one is fully satisfied by sensor feedback.

6.4 Description of Operation

When START is pressed, the system enters automatic mode. If the conveyor is free and sheets exist in the stack, the gripper moves above the stack, lowers, grips the sheet, lifts it, and moves to the conveyor. It lowers again, releases the sheet, returns to the top, and moves back to the stack to repeat the cycle. STOP or EMERGENCY cancels all steps and stops the system safely.